

## Problem Statement P05 · Cascading Failure

# When one bridge fails, what breaks next?

## DOMINO

Explainable Infrastructure Cascade & Intervention Engine

Find the next domino. Stop the cascade. Protect the dark zones.

### ONE BRIDGE, TRACED THROUGH OUR ENGINE



#### Bridge B07 fails

89% can still reach help



#### Coastal road N2

jumps to 1.3× capacity



#### Hospital & fire routes

87% of them use that road



#### Tower T-04 is down

no signal on the coast



#### Malpe ward, 5,200 people

no road, no signal: dark

TEAM NAME

**mossad.exe**

TRACK

**[Track]**

PROBLEM STATEMENT

**P05 · Cascading Failure**

SOLUTION TITLE

**DOMINO**

TEAM MEMBERS

[Member 1] · [Member 2] · [Member 3] · [Member 4] · [Member 5]

TODAY What gets affected?



DOMINO

Where can we intervene before the cascade grows?

ONE FAILURE, WALKED THROUGH BY OUR ENGINE

### 1 Bridge B07 fails

89% can still reach a hospital or fire station. On an ordinary damage map nothing looks wrong.

NOTHING VISIBLE YET

### 2 Traffic reroutes

Coastal road N2 is suddenly at 1.3× the traffic it was built for. That is the next domino.

DOMINO FOUND

### 3 Four actions tested

Three fail. Reopening a closed crossing holds the chain, and it takes 60 minutes.

CASCADE STOPPED

### 4 Now break B12

Same engine, different answer. Nothing holds it. 17,794 people lose their route to help.

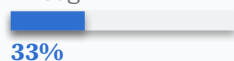
ONLY HARDENING HELPS

### Why that road, and not another

Overloaded right now



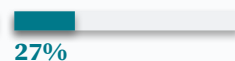
Network runs through it



Ambulance routes use it



People living behind it



### Dark zones: both, never either

No road to help AND no phone signal. Malpe goes dark at once, Kadiyali 25 minutes later.

$T_{\text{dark}} = \max(T_{\text{phys}}, T_{\text{comm}})$

THE ENGINE BEHIND IT, IN ORDER

Map the city



Break one asset



Reassign traffic



Score what breaks next

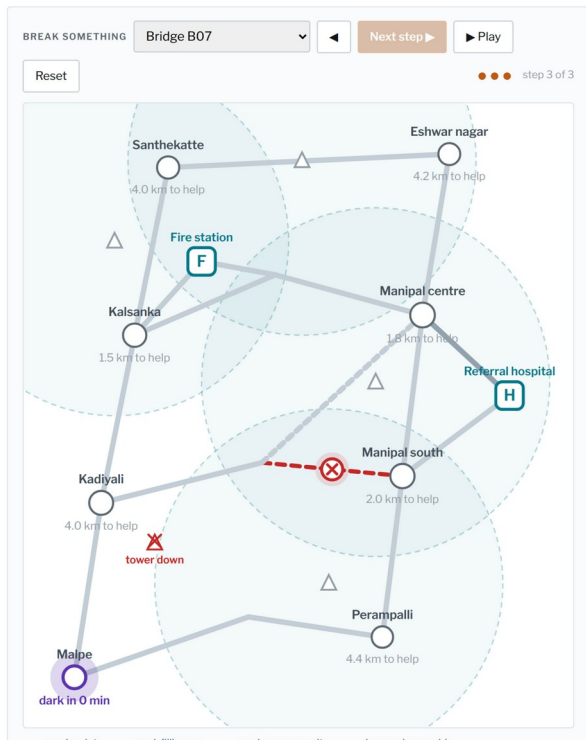


Test every action



Stop, or repeat

## OUR PROTOTYPE, RUNNING — NOT A MOCK-UP

**The chain stops here.**

Reopen Old crossing SR-2 works, and it can be done in 60 minutes. Access stays at 89%. Nobody is cut off.

PEOPLE WHO CAN STILL REACH A HOSPITAL OR FIRE STATION

**89.0%** everyone who could reach help before, still can start at 89%

WHAT BREAKS NEXT, AND WHY

**Coastal road N2**

It is now carrying 1.3x the traffic it was built for.

WHY THIS ONE, AND NOT ANOTHER ROAD

How overloaded it is right now 89%

How much of the network runs through it 33%

Whether ambulances and fire engines use it 87%

How many people live behind it 27%

WHAT WE TRIED, BEFORE IT BROKE

Contraflow on Coastal road N2  
can be done in 90 minutes NOT ENOUGH

Reinforce and widen Coastal road N2  
months of work, capital budget NOT ENOUGH

Reopen Old crossing SR-2  
can be done in 60 minutes WORKS

Reopen Old crossing SR-2 + contraflow on Coastal road N2  
can be done in 90 minutes WORKS

## THE STACK

**WorldPop + OpenStreetMap**  
data

**PostGIS**  
spatial store

**NetworkX + Dijkstra**  
routing and reassignment

**Domino detector**  
four-part score

**Intervention engine**  
counterfactual re-runs

**Dark-zone engine**  
isolation and countdowns

**React map and alerts**  
what an operator sees

**Real, and honestly scoped**

Real: routing, reassignment, the score, the counterfactuals, 17 failures and 5 budgets precomputed. Synthetic: the corridor itself. A live OSM extract replaces one loader function.

### EVIDENCE 1 — REAL POPULATION DATA

#### Remove a hospital, then put one action back

570,509 people in the study area, WorldPop plus mapped service locations



83.3% is above the 81.1% we started from, so the action that contains the cascade was worth taking anyway. 47.2% of the newly cut-off population is recovered, against 26.2% for random placement.

### EVIDENCE 2 — OUR ENGINE, RUNNING

#### Same budget, two ways to spend it

##### The standard method

reinforce the most central roads

1,993 people · ₹7.6 cr

##### DOMINO

buy back the most service per rupee

4,092 people · ₹8.9 cr

2.05× more people protected, for less money — and DOMINO stops at ₹8.9 crore because nothing else it could buy wins back another person. Corridor resilience goes 82.7 → 91.3 out of 100.

#### What ships first

Roads, bridges, services, towers and population on one graph, with the next-domino detector, the explainable score and the dark-zone clock.

#### What we do not claim

Access, not treatment. Topology and capacity, not human behaviour. Modelled population, not a building-level census. Costs are indicative rates.

#### How we will test it

Replay the Udupi–Manipal road closure of 20 May 2025 and check whether the roads DOMINO ranks next are the ones traffic actually moved onto.

**“We have a fixed budget. Where do we spend it?”**

#### WHAT WE SELL THEM

**A resilience score, per corridor**

**82.7 → 91.3** out of 100

#### WHO BUYS IT, AND HOW

Municipal authorities

District disaster management

Emergency services

Infrastructure operators

**B2G / B2B SaaS, plus deployment and integration**

#### ROADMAP, ONE LAYER AT A TIME



##### **Live traffic**

observed load, not modelled reroute



##### **Tower telemetry**

real coverage state during an event



##### **Peer-to-peer relay**

store-and-forward between nearby devices

#### THE ASK

One corridor, one monsoon season, with the Udupi city municipal council and the district disaster management authority, replaying last year's closures.

#### WHY THEY KEEP PAYING

The horizon changes every time the city builds something. A resilience score has to be recomputed each year, which turns a one-off disaster purchase into a line in the annual capital budget.

**Don't just predict the cascade.**

**Find where it can still be stopped.**

**DOMINO**